
Adversarial Attacks and Defenses for Traffic Sign Classification

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Abstract

Traffic sign recognition is an important component of autonomous driving, but image classifiers can be highly vulnerable to adversarial perturbations that are almost imperceptible to humans. In this project, we will study white-box untargeted adversarial attacks on a traffic sign classifier, evaluate robustness under standard and custom attacks, and compare several defense strategies including preprocessing, adversarial training, and denoising-based methods.

1 Introduction

Traffic sign classification is an important vision task in autonomous driving and driver assistance systems. If a classifier fails to recognize a traffic sign under adversarial perturbations, the resulting error may lead to unsafe decisions. For this reason, adversarial robustness is an important machine learning problem. The main idea of an adversarial attack is to add a small perturbation δ to an input image x so that the perturbed image $x_{\text{adv}} = x + \delta$ looks almost unchanged to humans but causes a classifier to make an incorrect prediction. In this project, we aim to better understand how adversarial attacks work and how to improve robustness against them in a practical traffic sign recognition setting.

The input to our algorithm is an image of a traffic sign, and the output is a predicted traffic sign class. We first train a baseline image classifier, then generate adversarial examples that attempt to maximize classification loss while keeping the perturbation bounded by a small ϵ . We focus on the white-box, untargeted setting. The white-box setting assumes that the attacker knows the model parameters and gradients, which gives a strong and transparent robustness evaluation. The untargeted setting is chosen because it is a natural first step: the goal is simply to cause any misclassification rather than forcing a specific target label. After evaluating the baseline model, we will compare multiple defense methods to determine which strategies improve robustness most effectively.

2 Dataset and Features

Our main dataset will be the German Traffic Sign Recognition Benchmark (GTSRB) (1). We choose this dataset because it is directly related to autonomous driving and provides a more realistic benchmark for this project than simpler datasets such as MNIST. If computation time or implementation difficulty becomes an issue, we may use CIFAR-10 or MNIST for preliminary experiments or debugging. The input features are the traffic sign images themselves. We will resize images to a fixed resolution, normalize pixel values, and optionally apply simple augmentation during training.

Our baseline classifier will be ResNet-18 (2), although a smaller CNN may be used in early experiments for faster iteration. The model takes an image x as input and outputs a probability distribution over traffic sign classes. For robustness evaluation, we will generate adversarial versions of the validation/test set using several attack methods and compare classifier performance across clean and adversarial inputs. The main features used by the model are the visual patterns in the image, including color, shape, texture, and edge information. Since traffic sign recognition depends heavily on shape and boundary cues, we are particularly interested in edge-focused attacks and defenses.

3 Methods

We will first train a baseline classifier on clean images using cross-entropy loss and standard gradient-based optimization. After obtaining a clean baseline, we will generate adversarial examples in the white-box untargeted setting using several attack methods. The first is FGSM (3), which perturbs the input in the sign direction of the input gradient:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x L(f_\theta(x), y)).$$

The second is PGD (4), which is a stronger iterative attack that repeatedly updates the input while projecting it back into the allowed ϵ -ball:

$$x^{t+1} = \Pi_{B_\epsilon(x)}(x^t + \alpha \cdot \text{sign}(\nabla_x L(f_\theta(x^t), y))).$$

We will also evaluate the Carlini & Wagner (C&W) attack (5), an optimization-based attack that searches for a small perturbation that still causes misclassification. In addition, we propose a custom *edge-aware attack*, motivated by the intuition that traffic sign classification

depends strongly on edge and shape information. Instead of perturbing all pixels uniformly, this method will focus perturbations on edge regions and test whether such localized perturbations are especially effective.

We will evaluate the baseline model using clean accuracy, adversarial accuracy, attack success rate, and perturbation cost. Clean accuracy is the accuracy on the original test set. Adversarial accuracy is the accuracy on the attacked test set. Attack success rate measures the fraction of samples that are correctly classified in the clean setting but misclassified after attack. Perturbation cost measures how much the adversarial example differs from the clean image, for example under an L_∞ -bounded perturbation budget. For defense methods, we will compare: (1) no defense, (2) preprocessing-based defenses such as Gaussian blur, median filtering, and JPEG compression, (3) adversarial training, and (4) custom denoising methods based on morphology and the Fourier domain. In adversarial training, for each mini-batch, we generate adversarial examples using the current model and update the model parameters using the adversarial loss, optionally together with clean loss. This can be viewed as the robust optimization problem

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\| \leq \epsilon} L(f_{\theta}(x + \delta), y) \right].$$

Finally, after training the baseline and defense models, we will use AutoAttack (6) as a final robustness benchmark to test whether the models remain effective under stronger standardized attacks.

4 Deliverables

These deliverables are grouped by priority. Our must-accomplish goals define the minimum requirements for a successful project, while the later categories describe extensions that we expect or hope to complete if time permits.

4.1 Must accomplish

1. Train a baseline traffic sign classifier on GTSRB and report clean accuracy.
2. Implement and evaluate standard adversarial attacks such as FGSM and PGD.
3. Compare at least two defense methods and report clean accuracy, adversarial accuracy, attack success rate, and perturbation cost.

4.2 Expect to accomplish

1. Implement the C&W attack and compare it with FGSM and PGD.
2. Design and evaluate the custom edge-aware attack.
3. Compare preprocessing-based defenses with adversarial training on the same evaluation protocol.

4.3 Would like to accomplish

1. Evaluate custom denoising defenses based on morphological operators and FFT-based filtering.
2. Apply AutoAttack for a final robustness comparison across the baseline and defense models.
3. Analyze which attack-defense combinations are most effective for traffic sign classification and discuss why.

References

References

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